

Stage 1.2 — Non-Constant Isp Ascent (Hard Mode)

Era: Redstone/Mercury-Redstone class, ~1961 (fictional program), continuing Project Arrowhead. **Target time:** a full afternoon. **Prerequisite:** Stage 1.0 (Range Table Targeting), Stage 1.1 (Liftoff Check & Gravity-Turn Loss).

Why this problem exists

1.0 treated specific impulse as a single constant, 235 s, for the whole burn. Real engines don't work that way: thrust is $F = \dot{m} \cdot V_e + (P_e - P_{amb}) \cdot A_e$ — as the vehicle climbs and ambient pressure P_{amb} drops, the same engine produces more thrust and a higher effective Isp, right up to a vacuum ceiling. This is standard, period-authentic propulsion analysis (see Sutton, *Rocket Propulsion Elements* — in print and in active engineering use well before 1961) and it is **not** a small effect: most of the pressure drop happens in the first ~100,000–150,000 ft, which is most of this vehicle's burn. Treating Isp as constant, as 1.0 did, understates burnout velocity by a meaningful margin — you'll see by how much below.

This is genuinely harder than 1.0, for a specific reason: Isp now depends on altitude, altitude depends on the velocity history, and velocity depends on Isp. That circular dependency is why this can't be solved in one closed-form step — it has to be marched forward in time, a handful of steps at a time, the same way early trajectory integrators (Gauss-Jackson and similar — already flagged in the campaign's reference list as a harder future problem in its own right) worked before continuous integration was practical by hand. What you're doing here is a simplified version of that same idea.

Given data

Same (corrected) vehicle as 1.0/1.1: $W_0 = 55,000$ lbf, $W_p = 40,000$ lbf burned uniformly over $t_b = 143$ s (weight decreases linearly with time). Standard gravity $g_0 = 32.174$ ft/s².

Loss budget: $\Delta v_{\text{losses}} = 3540$ ft/s, from Stage 1.1's first-principles gravity-turn/drag study.

Engine performance — Isp vs. altitude (from separate propulsion analysis; reflects the nozzle's altitude compensation as ambient pressure drops):

	h (ft)	Isp (s)
0	235	
20,000	246	
40,000	253	
60,000	258	
80,000	261	
100,000	263	
120,000	264	

h (ft)	Isp (s)
$\geq 140,000$	265 (vacuum plateau)

Ascent altitude profile: reuse the same trajectory checkpoints from Stage 1.1 (same vehicle, same preliminary shaping run) rather than a separately-invented profile:

t (s)	h (ft)
0	0
13	604
39	6,485
65	19,230
91	37,878
117	63,427
143	100,235

Explicit simplification, stated up front: this problem does *not* solve for the trajectory shape — the altitude profile above is a given input (1.1’s result), not something you derive from the velocity you’re computing here. That keeps the coupling one-directional (altitude $\rightarrow$ Isp $\rightarrow \Delta v$) and hand-tractable. The full version, where Isp-vs-altitude feeds back into the trajectory shape itself, is real and harder — a natural candidate for a future stage, not this one.

Authentic method + reference material

Divide the burn into the six intervals defined by the checkpoints above (note: unequal step sizes — the first is 13 s, the rest are 26 s each, since they come from 1.1’s checkpoint times, not an even split). For each step:

1. Look up altitude at the START of the step (from the given profile).
2. Interpolate Isp at that altitude.
3. $c = Isp \cdot g_0$
4. $\Delta v_{\text{step}} = c \cdot \ln(W_{\text{start}}/W_{\text{end}})$ [W_end from the linear weight burn]
5. Accumulate: $V_{\text{cumulative}} += \Delta v_{\text{step}}$

Note the numerical choice being made in step 1: using the altitude at the *start* of each step to look up Isp for that whole step is an explicit (forward-lagged) scheme — it’s simple and hand-tractable, but it means each step’s Isp is slightly stale (the vehicle is actually higher, and Isp actually slightly better, by the step’s end). A more accurate scheme would use the midpoint or average altitude of the step. This is the same kind of approximation-choice tradeoff numerical integrators always face — flagged here so it’s a visible decision, not a hidden one.

Worksheet

Step	t_start-t_end (s)	h_start (ft)	Isp (interp.)	c = Isp·g ₀	W_start	W_end	ln(W_start/W_end)	Δv_step (ft/s)	V_cumulative (ft/s)
1	0-13	0			55,000	51,364			
2	13-39	604			51,364	44,091			
3	39-65	6,485			44,091	36,818			
4	65-91	19,230			36,818	29,546			
5	91-117	37,878			29,546	22,273			
6	117-143	63,427			22,273	15,000			

Total ideal Δv (altitude-varying Isp) = _____ ft/s

V_{bo} = Total ideal Δv – 3540 ft/s = _____ ft/s

Compare to 1.0's constant-Isp result (V_{bo} ≈ 6284 ft/s): difference = _____ ft/s (_____ %)